

Rajat

Chemical Engineering (B.Tech.)

Gender: Male

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Examination	Degree	CGPA	University	Year
Under Graduate	B. Tech.	8.15	NIT Jalandhar	2027

EXPERIENCE

Defence Research & Development Organisation (DRDO)

[Jun'26-Jul'26]

- Built a **physics-informed ML pipeline** to screen 25+ candidate material formulations, engineering features (mass-loss, enthalpy, decomposition kinetics) from thermogravimetric (TGA) analysis data using Python and Proteus thermal-analysis software.
- Trained **XGBoost surrogate models** ($R^2 \approx 0.88-0.89$) on simulated thermal-decomposition profiles to predict material performance, reducing dependence on costly physical testing cycles.
- Ran **Bayesian optimization (Optuna)** across a 50,000-scenario Latin-Hypercube-sampled search space to identify optimal multi-material blend configurations, extracting a **Pareto front** of cost/performance trade-offs to standardize final decision-making.
- Validated model predictions against 200+ real experimental data points and stress-tested shortlisted candidates for $\pm 2\%$ robustness before sign-off.
- Built **Streamlit dashboards** to visualize decomposition curves and model outputs for senior DRDO scientists; findings fed into a formal presentation and a filed patent.

PROJECTS

CampusBite — NIT Jalandhar Canteen Hub | [GitHub](#)

[Jun'25]

- Built a **full-stack web application** (React.js, TypeScript, Node.js, Express.js) that unifies 6 campus canteens on one platform, showing menus, operating status, capacity, live crowd level and seat availability.
- Implemented **seat booking and cancellation** with capacity-aware validation, **REST APIs** with type-safe contracts and **role-based access control** (student/admin); persisted canteens, menus, bookings and occupancy in a **MySQL-compatible database** using **Drizzle ORM**.
- Developed an **explainable ML model** (Python, scikit-learn, Pandas, NumPy) to predict canteen traffic (low / moderate / high / full) from day of week, time of day, meal period, historical occupancy, existing bookings and capacity; surfaced predicted peak hours and less-crowded recommendations.
- Designed a **responsive, mobile-first UI** with reusable components, a student dashboard (occupancy, bookings, history) and an admin panel; documented setup with seeded demo data and Git-based version control.

Crude Oil Market Analytics Pipeline | [GitHub](#)

[Apr'26]

- Designed and deployed an end-to-end data/ML pipeline — **EIA API** → **Airflow orchestration** → **Azure Blob Storage raw landing zone** → **Postgres** → **dbt transformations** → **Streamlit dashboards** — containerized with Docker; provisioned resource groups, storage accounts and containers via **Azure CLI**.
- Implemented incremental, idempotent ingestion in Python (requests, psycpg2) with automatic backfill and upsert logic keyed on max-loaded period per series.
- Modeled staging → intermediate → mart layers in dbt with built-in data-quality tests, cutting manual QA effort and improving data reliability for downstream dashboards.

Local Multimodal LLM Agent Pipeline (ASR → LLM → TTS + Browser Automation) | [GitHub](#)

[May'2026]

- Built a local, privacy-preserving voice agent combining **speech-to-text (Whisper)**, a **locally-hosted LLM (Ollama / Llama 3.1)** for reasoning, and **text-to-speech (pyttsx3)** for response generation.
- Integrated **Playwright-based browser automation** so the agent could execute multi-step web tasks from natural-language voice commands, exploring practical agentic-AI orchestration end to end.

SKILLS

Machine Learning: XGBoost, scikit-learn, Pandas, NumPy, Bayesian Optimization (Optuna), surrogate modeling, predictive modeling, feature engineering

Web Development: React.js, TypeScript, Node.js, Express.js, HTML5, CSS3, REST APIs, Drizzle ORM, MySQL, role-based access control

Data Engineering / MLOps: Airflow, dbt, Docker, PostgreSQL, Azure (Blob Storage, CLI resource provisioning), Git/GitHub, Streamlit

GenAI / Agentic AI: LLM integration (Ollama/Llama 3.1), Whisper speech-to-text, agent orchestration, browser automation (Playwright)

Languages & Tools: Python, SQL, TypeScript, MATLAB, Power BI, Microsoft Office

POSITION OF RESPONSIBILITY

- ChESS (Chemical Engineering Students' Society), NIT Jalandhar** – Executive Member: Coordinated the "Chem Hunt" event end-to-end, managing logistics, timelines, and stakeholder buy-in across teams for smooth execution [Oct'24 - Present]
- CHEM-CON, Technical Team (Management)** – Managed cross-functional coordination for ChemCon 2024, ensuring smooth technical operations and vendor/stakeholder support. [Dec'24]
- Think India, NITJ** – Active Volunteer: Organised and led a tribute event (Black Day, Feb 14) involving multi-stakeholder coordination across content, logistics, and participation. [Nov'24 - Present]
Coordinated and managed an awareness event on "One Nation, One Election," handling event planning, participant coordination, and on-ground execution. [Oct'25]

EXTRA CURRICULAR ACTIVITIES

- 3rd Place, Case Study Challenge – EXERGY, IIT Kanpur (Chemineers Society)
- Runner-up, Sudo Hackathon – DAVIET

[March'24]

[May'25]